Understanding **AI Engineering**

DELL Technologies

About Me



- Member of the AI Innovation Hub (AIH) team in Dell Technologies
- 20+ years as an IT Practitioner
- Software Development, Project Management, Operations Management, DevOps, Pre-Sales and Post-Sales
- Passionate about Cloud Native technologies and their emerging trends

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What exactly is AI Engineering?

The discipline of **building, deploying, and maintaining** AI-powered systems in the real world.



AI Research

Discovering what's possible



Software Engineering

Building things that work reliably



Domain Expertise

Knowing what problems actually matter

AI Engineers don't just train models. They **design systems, connect models, evaluate performance,** and ensure **safety and responsibility.**

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Why AI Engineering Matters Now

AI is everywhere
But most of it is built badly or irresponsibly

The skills gap is massive
Companies are desperate for people who can actually implement AI solutions

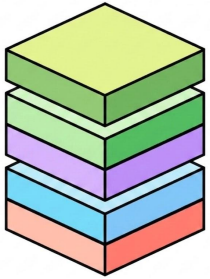
Tools are more accessible
You don't need a supercomputer to build something impressive

Mistakes are consequential
Bad AI in production can cause real harm

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The Full Stack of AI Engineering

Layer	What It Covers
Data	Collecting, cleaning, and preparing information
Models	Training, fine-tuning or using pre-built AI models
Applications	Connecting models to real products
Evaluation	Testing whether it actually works
Security	Making sure it can't be abused



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Getting Started in AI Engineering

Learning Objectives:

- Understand core AI concepts and how modern AI system work
- Gain hands-on experience with LLM and AI tools
- Learn fundamental practices for building safe and responsible AI applications



Topics	
H1 2026	**H2 2026**
• Introduction to AI, ML and Deep Learning	• Generative AI Applications
• Natural Language Processing	• Agentic AI and LLM Agents
• LLM Fundamentals	• Model Context Protocols (MCP)
• Working with Pre-trained LLMs	• Evaluations, Ethics and Responsible AI
• AI Security in LLM	• AI Production Deployments

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AI, Machine Learning & Deep Learning

The Bedrock of Modern AI

Before we can work with modern AI, we need to understand how machines actually learn.


Artificial Intelligence: The broad science of mimicking human abilities.


Machine Learning: Algorithms that improve through experience.


Deep Learning: Multi-layered neural networks for complex patterns.

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AI, Machine Learning & Deep Learning

What you'll learn

- What makes something "AI" vs. a regular program
- How machines learn from data (supervised learning, classification)
- What a neural network actually is — layer by layer
- Deep learning and why images changed everything

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Natural Language Processing

Teaching Machines to Read

Natural Language Processing is the field of AI that deals with human language — reading it, understanding it, and generating it.



Why it matters

Most of the world's information exists as text. If a machine can read and understand text, it can do almost anything.

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Natural Language Processing

What you'll learn



How computers represent words as numbers



Tokenisation — breaking text into meaningful chunks



Text classification — "is this email spam?"



Word embeddings — how words gain meaning in context



Building toward modern language models

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Large Language Models

What changed?

In 2017, the **Transformer architecture** was introduced in the paper *"Attention Is All You Need"*. It completely rewrote the rules for AI.

What makes LLMs different:

Massive Training Data

Trained on vast amounts of text including books, websites, code, and conversations.

Relationship Modeling

They don't just match patterns—they model complex relationships between ideas.

Contextual Generation

Ability to generate coherent, contextual text at scale across various domains.

Natural Prompting

Interact using plain English—no special syntax or programming knowledge needed.

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How LLMs Work?

Step 1 — Tokenisation

Text is broken into tokens (chunks of words). "Unbelievable" becomes ["Un", "believ", "able"]. Each token gets a number.

Step 2 — Embeddings

Numbers convert into vectors (hundreds of numbers) capturing meaning. Similar meanings cluster together.

Step 3 — Attention

The model asks: "which tokens should influence others?" like "bank" looking at "river" to understand context.

Step 4 — Prediction

The model picks the most likely next token, appends it, and repeats — generating text one token at a time.

Step 5 — Training

Billions of text examples were used to adjust internal weights. After enough iterations, an LLM emerges.

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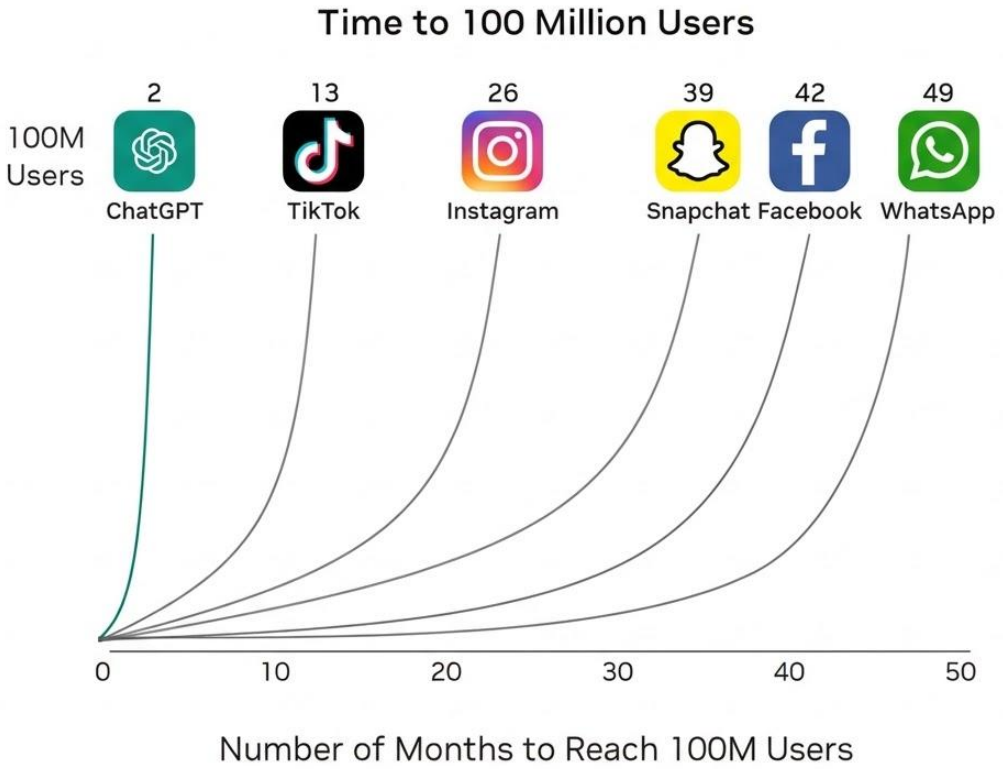
Large Language Models (LLMs)

What you'll learn

- How to interact with LLMs effectively
- Prompt engineering fundamentals
- What LLMs can and can't do

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Generative AI



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Generative AI

- Most early AI was built to classify or predict — *is this email spam? What's the price tomorrow?*
- Generative AI does something fundamentally different: it creates new content that didn't exist before.
- What can it generate?
  - 📝 Text — essays, code, summaries, conversations
  - 🖼️ Images — from a text description to a finished picture
  - 🎵 Audio — music, voice, sound effects
  - 🎬 Video — synthetic footage, animations
  - 💻 Code — entire functions, scripts, applications

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Generative AI

What's driving it?

The same Transformer architecture behind LLMs — scaled up, trained on multimodal data, and fine-tuned for generation rather than just prediction.

The key shift in thinking:

✕

You're no longer programming a computer to follow instructions step by step.

✦

You're describing what you want — and the model figures out how to produce it.

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Agentic AI

What is an AI Agent?

An agent is an AI system that doesn't just respond to a question — it plans and executes a sequence of actions to achieve a goal.

Chatbot vs. AI Agent

Chatbot

- Responds to one prompt
- Stateless
- Reads only
- One-shot

AI Agent

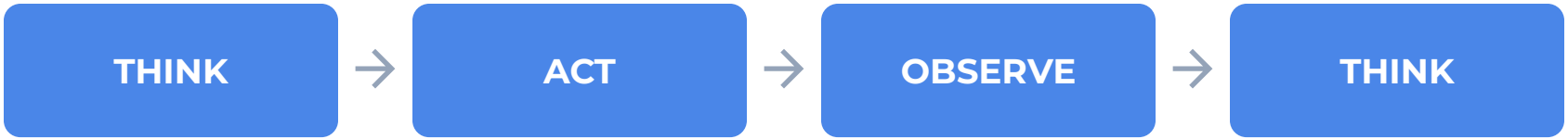
- Plans across multiple steps
- Maintains memory and state
- Can use tools (search, code, APIs)
- Loops until the task is done

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How Agents Think and Act

The ReAct Loop (Reason + Act)

Every LLM agent operates in a continuous loop of thinking and acting until a goal is achieved.



repeat

Crucial Concepts

- **Tools:** External functions like search or APIs
- **Memory:** Short & long-term state
- **Condition:** How it knows it's done

Breaking it Down

The LLM reasons (THINK), calls a tool (ACT), reads results (OBSERVE), and updates its plan (THINK AGAIN).

Risks & Safety

- Infinite loops
- Indirect prompt injection
- Irreversible actions

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Agentic AI

What you'll learn

- Agents that can search the web, run code, and call APIs
- Multi-step reasoning pipelines
- Systems with memory that persists across interactions

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Evaluation, Ethics & Responsible AI

The Uncomfortable Truth

Most AI systems are deployed before anyone properly checks if they work well — or if they cause harm.

Evaluation

Is my model accurate? On what inputs does it fail?

Is it consistent across all users?

How to measure quality without one right answer?

Ethics

Who is harmed if this goes wrong?

Does it treat all groups of people fairly?

Am I transparent about its limitations?

In Practice

Model cards & docs

Bias audits

Red-teaming

Human-in-the-loop design

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Evaluating Models

The Core Principle
Crucial and integral part of AI Engineering to ensure robust and reliable systems.



Risk Mitigation

Mitigate risks and uncover opportunities by identifying potential failure modes early in the development lifecycle.



Deployment Readiness

Determine application usefulness and readiness for deployment through rigorous performance benchmarking.



Continuous Improvement

Foster continuous improvement of models and applications based on iterative evaluation feedback loops.

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Challenges of Evaluating AI Systems

Natural Complexity

The challenges tackled by these models are often naturally complex.

Open-Ended Tasks

Tasks are usually **open-ended**, with multiple valid answers.

Black Box Nature

Models operate as **black boxes**, making inner workings hard to interpret.

Benchmark Saturation

Public benchmarks become **saturated** quickly as models achieve perfect scores.

Emerging Capabilities

Crucial to evaluate existing tasks and uncover new, emerging capabilities.

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What Happens When AI Goes Wrong?

The Takeaway
These aren't edge cases. They're predictable failure modes of systems built without proper evaluation and safeguards.

Hallucination

LLMs confidently state things that are factually wrong. A model told a lawyer to cite six non-existent case precedents used in court.

Bias

A hiring algorithm learned to downgrade résumés from women, reflecting and amplifying past discrimination.

Prompt Injection

Malicious instructions hidden in summaries can force AI to forward private emails to attackers.

Over-reliance

Automation bias causes pilots or doctors to trust AI results without checking, leading to critical errors.

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AI Security in LLMs

A New Attack Surface

LLMs introduced vulnerabilities that traditional security tools weren't designed to catch. The inputs are natural language. The outputs are unpredictable. The attack vectors are creative.



Direct Prompt Injection

Crafting inputs that override the model's instructions



Indirect Prompt Injection

Hiding malicious instructions in content the agent retrieves



Model Poisoning

Corrupting a model during training with malicious data



RAG Poisoning

Injecting bad content into the knowledge base an LLM draws from



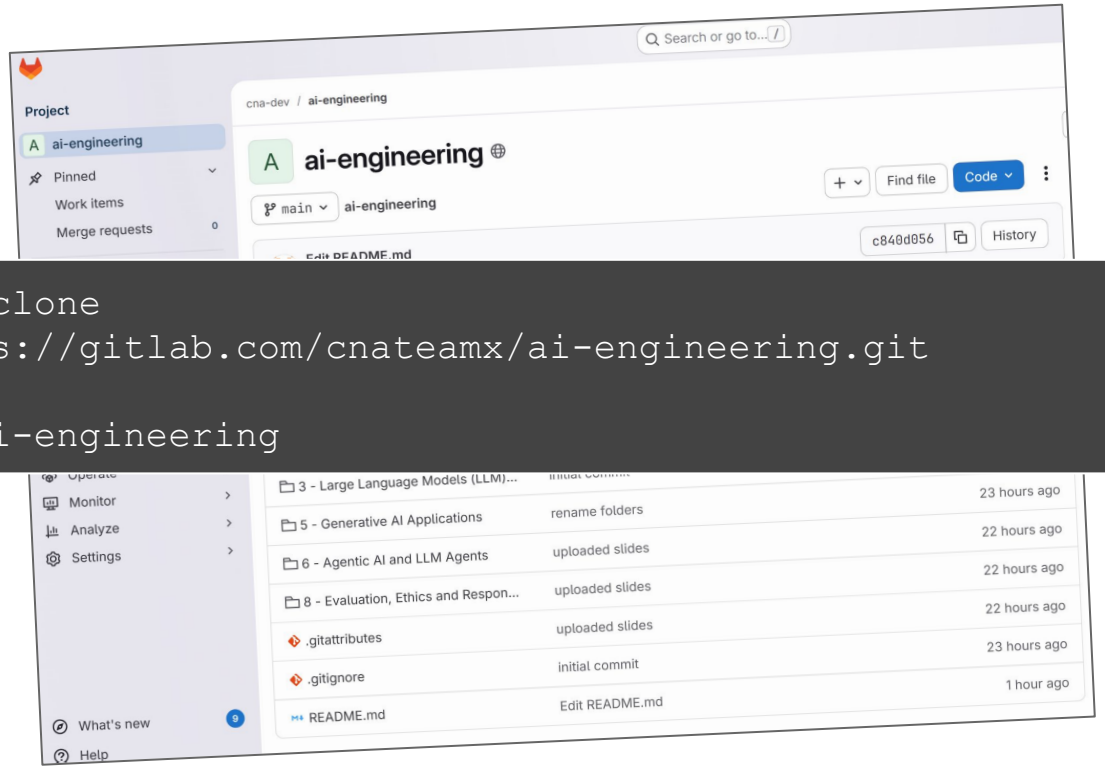
Agentic Attacks

Exploiting the autonomy of AI agents to cause harm

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Accessing the Labs

Scan me!




```
git clone
https://gitlab.com/cnateamx/ai-engineering.git

cd ai-engineering
```

https://gitlab.com/cnateamx/ai-engineering

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What You'll Need to Get Started

The essentials: Prerequisites and tools for your AI Engineering learning.

Python 3.9+
The primary language for all labs and scripts.

Jupyter
Your interactive lab environment (Lab or Notebook).

pip
Install packages via pip install -r requirements.txt.

Docker Desktop
Essential for the **Topic 10 AI Security** labs.

API Key
Required for LLM calls; configure using the .env.example file.

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How the Labs Are Structured

Interactive Environment
Everything runs in **Jupyter Notebooks**: interactive documents that combine text, code, and output.

Learn

Read detailed explanations and theoretical context behind each concept.

Examine

See clear code examples illustrating the implementation of AI techniques.

Execute

Run the code yourself and visualize the results immediately within the document.

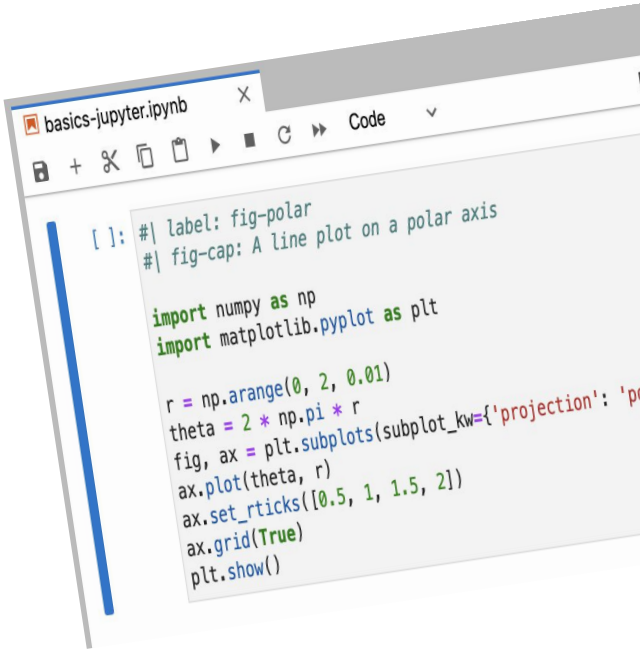


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Accessing the Labs

How each lab is structured:

1. 📖 **Read the context** — markdown cells explain what you're doing and why
2. 👀 **Study the code** — understand it before running it
3. ▶ **Run the cells in order** — top to bottom, always
4. 🔧 **Experiment** — change values, break things, see what happens
5. 💬 **Reflect** — what did you observe? What surprised you?



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What Good AI Engineering Looks Like

Start with the Problem

Focus on user needs, not just using an LLM. Is AI the right tool for the job?

Evaluate Rigorously

Test on diverse inputs, probe failure modes, and document limitations.

Design for Failure

Plan for wrong outputs with fallbacks, oversight, and graceful degradation.

Stay Curious

Master fundamentals like data, evaluation, and security in a fast-moving field.

Take Responsibility

AI harm isn't the algorithm's fault. Engineers own the deployment and testing.

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Moving from traditional ML Engineering

Evolution of the Field

Significant shift in focus moving from traditional Machine Learning (ML) Engineering towards AI Engineering.

ML Engineering

Focuses on developing machine learning models ***from scratch***.

AI Engineering

Builds on top of ***existing models*** to create intelligent applications.

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